

POCKETSMART AI

System Architecture

1. Introduction

The system architecture describes the major components of PocketSmart AI and how they communicate with each other.

2. Architecture Description

The user interacts with PocketSmart AI through a web browser. The request is received by the FastAPI backend. The backend validates the input and sends the required information to the appropriate application service.

For AI-based requests, the application creates a suitable prompt and sends it to the Generative AI service. The AI service processes the prompt and returns a generated response. The application formats the response and displays the result to the user.

3. Main Components

User Interface:

The user interface allows users to enter their budget, room, style, and required items.

FastAPI Backend:

The FastAPI backend manages application requests and connects the frontend with the application services.

Authentication Module:

The authentication module manages registration and login functionality.

Planner Module:

The planner module processes user planning information.

AI Service:

The AI service communicates with the Generative AI API and processes AI-generated responses.

Database:

The database stores required application and user information.

Result Display:

The generated plan is returned to the frontend and displayed to the user.

4. Architecture Flow

**User → Web Browser → FastAPI Backend → Application Service → Generative AI API
→ AI Response → Backend → Web Browser → User**

